

AI-Powered Autonomous Building Management System

Integrating EnergyPlus with Open-Source LLMs using MCP

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1. Executive Summary

Buildings consume nearly 40% of global energy, making intelligent energy management a critical challenge. This project presents an AI-powered Autonomous Building Management System (AI-BMS) integrating EnergyPlus, Qwen3-8B, and the Model Context Protocol (MCP). The framework continuously monitors building conditions, constructs a structured prompt from operational context and historical memory, reasons over multiple objectives, validates the generated recommendations, and executes safe control actions through EnergyPlus actuators to form a closed-loop optimization system.

2. Problem Statement

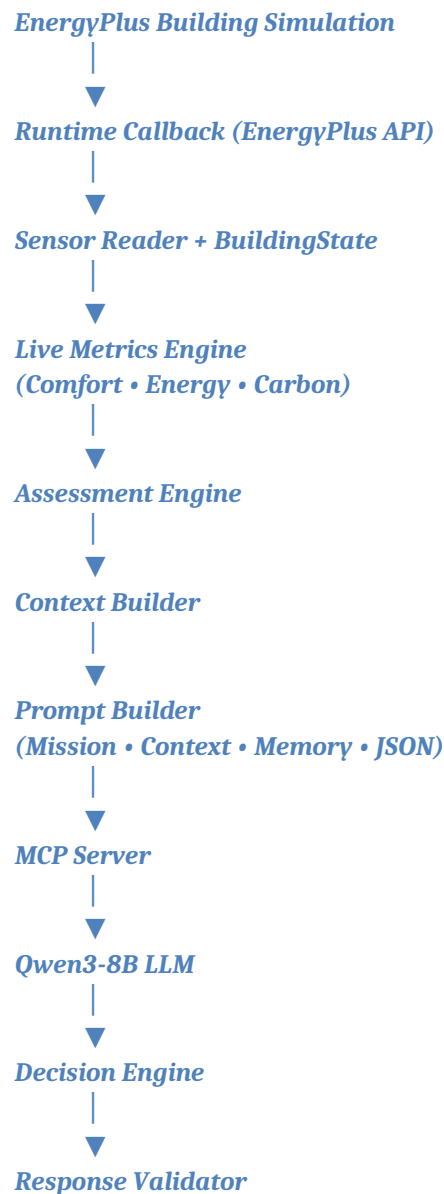
Conventional Building Management Systems rely on fixed rules and thresholds. They cannot jointly optimize energy efficiency, occupant comfort, and carbon emissions under changing operating conditions. The objective of this project is to develop an AI-driven supervisory

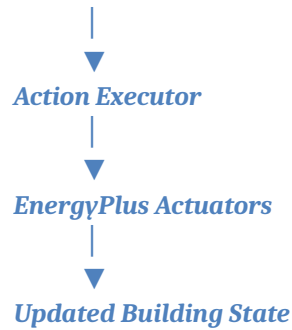
controller capable of understanding the building state, reasoning over competing objectives, and autonomously updating operating parameters.

3. Proposed Solution

The proposed solution is organized into six layers: (1) Building Simulation, (2) Runtime Data Acquisition, (3) Building Intelligence, (4) Prompt Engineering, (5) LLM Decision Making, and (6) Autonomous Execution. Each layer performs a dedicated responsibility while exchanging structured data with adjacent layers.

4. Overall System Architecture

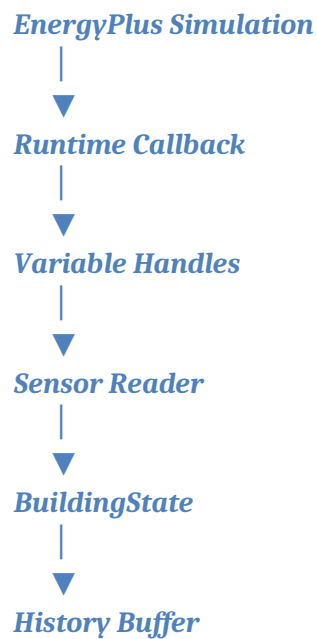




Architecture Explanation

The architecture follows a layered design. EnergyPlus performs the physical simulation of the building. Runtime callbacks acquire live sensor values and convert them into a standardized BuildingState object. The Live Metrics Engine evaluates thermal comfort, energy consumption, and carbon emissions. The Context Builder combines current conditions, historical trends, objectives, and constraints into a unified Context object. The Prompt Builder transforms this context into a structured prompt for the Qwen3-8B language model via MCP. The model returns optimization recommendations that are validated before the Action Executor applies safe control actions back to EnergyPlus, completing the feedback loop.

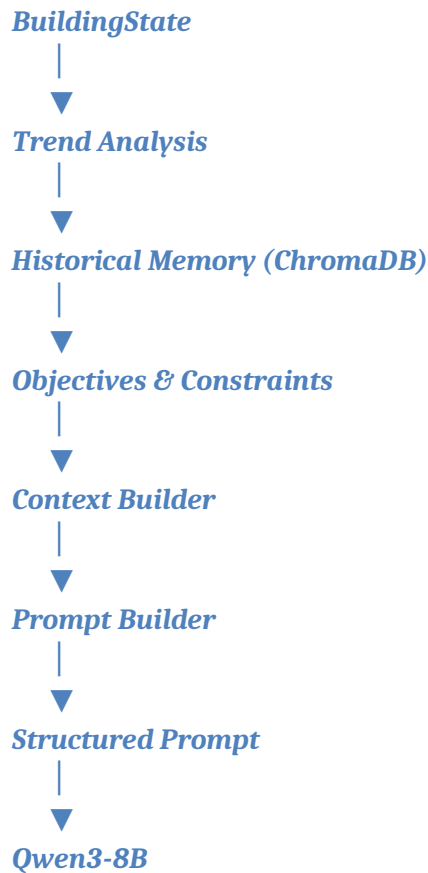
5. EnergyPlus Runtime Architecture



Explanation

The EnergyPlus Runtime API invokes Python callback functions at each simulation timestep. Variable handles are used to retrieve weather, HVAC, lighting, occupancy, and zone information. These values are encapsulated within the BuildingState model and stored in a history buffer for temporal analysis and episodic memory retrieval.

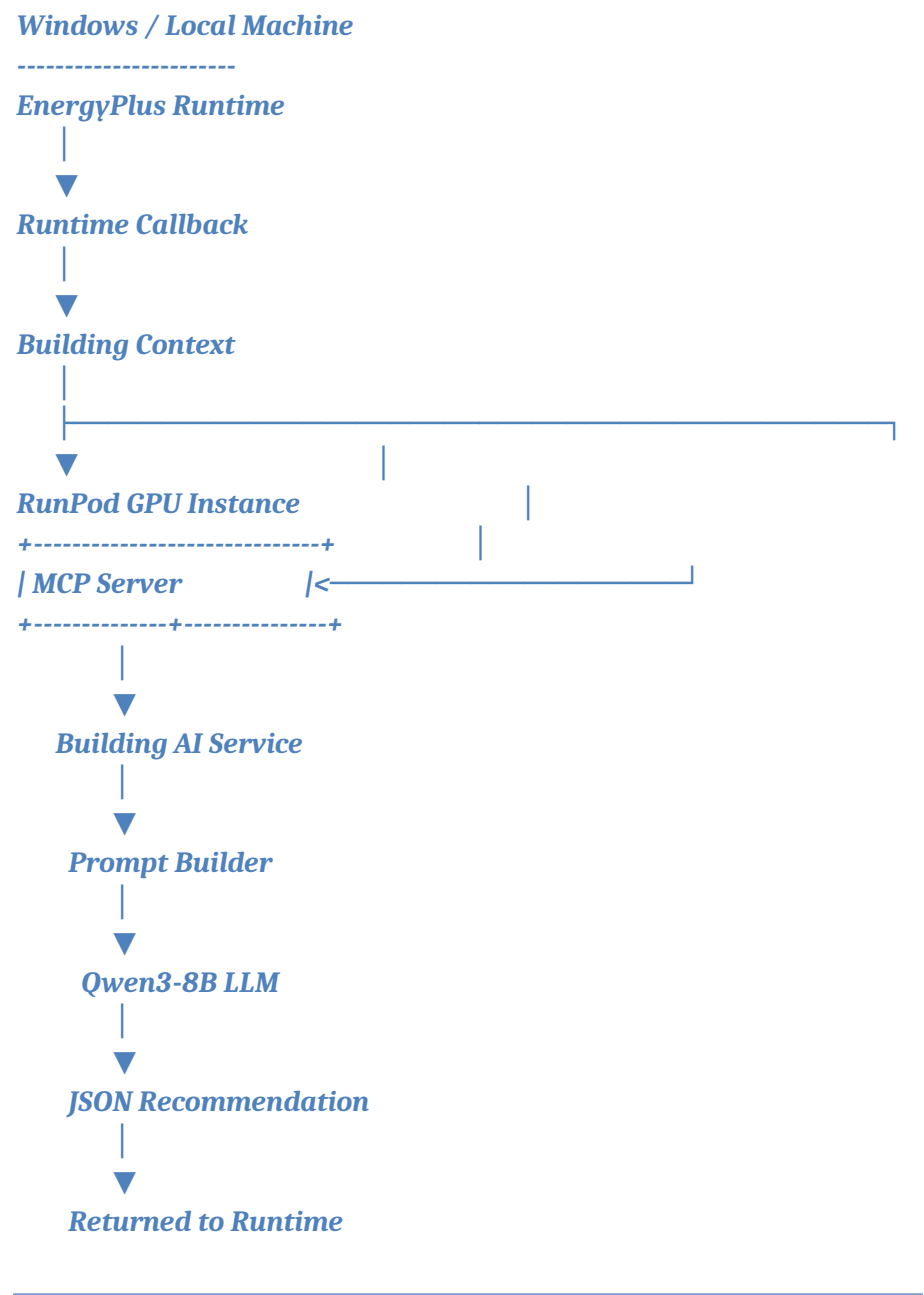
6. Prompt Engineering



Prompt Construction Strategy

The prompt is engineered rather than generated from raw sensor values. It contains six sections: Mission, Current Building Context, Performance Metrics, Historical Episodic Memory, Operational Constraints, and Required JSON Output Schema. This structure enables deterministic, explainable, and machine-readable responses that can be validated and executed automatically.

7. MCP Integration Architecture

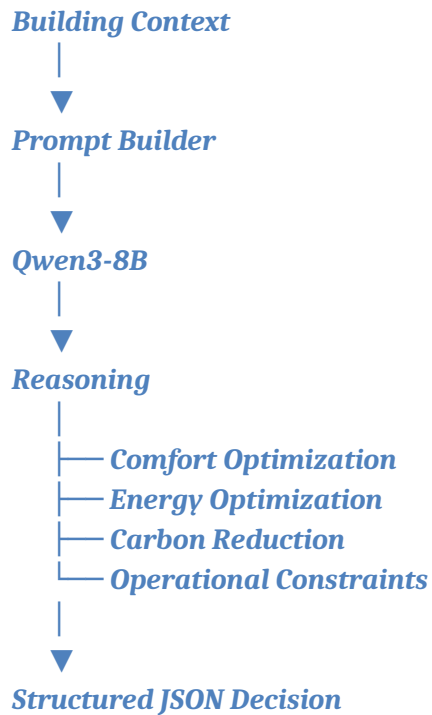


Explanation

The Model Context Protocol (MCP) acts as the communication layer between the EnergyPlus runtime and the AI reasoning engine. Runtime data is packaged into a context object and sent to the MCP server, which invokes the Building AI Service. The Prompt Builder constructs the final prompt, the Qwen3-8B model generates structured

recommendations, and validated JSON decisions are returned to the local runtime for execution.

8. LLM Reasoning Pipeline



Explanation

The language model receives a structured prompt instead of raw telemetry. It reasons over multiple competing objectives and returns a machine-readable JSON response containing recommended actions, rationale, confidence, and expected outcomes.

9. Closed-Loop Evaluation

EnergyPlus



Read Sensors



BuildingState



Metrics & Assessment



Context Builder



Prompt Builder



Qwen3-8B



Decision Validation



Action Executor



EnergyPlus Actuators

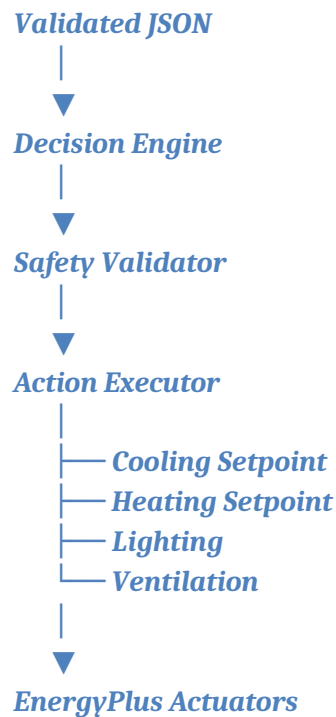


Next Simulation Timestep

Explanation

At every simulation timestep the controller collects sensor values, evaluates building performance, generates a structured prompt, obtains AI recommendations, validates them against engineering constraints, executes approved actions through EnergyPlus actuators, and repeats the process. This feedback loop enables autonomous supervisory control.

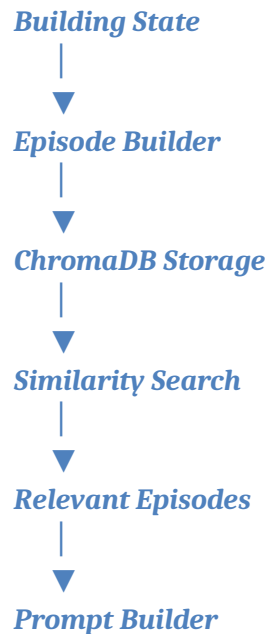
10. Action Execution Pipeline



Explanation

Only validated recommendations are translated into actuator commands. This separation between reasoning and execution prevents unsafe operating conditions and ensures the AI operates within engineering limits.

11. Episodic Memory (ChromaDB)



Explanation

Each control cycle is stored as an episode containing building conditions, decisions and outcomes. Similar historical episodes are retrieved and injected into future prompts, allowing the model to reason with prior operational experience.

12. Dashboard

The dashboard presents:

- Indoor and outdoor temperatures
- HVAC and lighting energy consumption
- Thermal comfort (PMV/PPD)
- Carbon emissions
- AI recommendations
- Executed actions
- Historical trends and savings
- System health and runtime status

13. Technology Stack

EnergyPlus, OpenStudio, Python, Qwen3-8B, Hugging Face Transformers, Model Context Protocol (MCP), ChromaDB, PyTorch, Streamlit, GitHub.

14. Deliverables

- Source code
- EnergyPlus building model
- MCP server
- AI reasoning pipeline
- Prompt engineering framework
- Closed-loop controller
- Dashboard
- Architecture documentation
- Demonstration workflow

15. Future Enhancements

Future work includes reinforcement learning for adaptive optimization, multi-building coordination, electricity price-aware scheduling, renewable energy integration, predictive occupancy modeling, and deployment on physical Building Management Systems.

16. Conclusion

The proposed AI-powered Building Management System demonstrates how EnergyPlus, the Model Context Protocol, and an open-source LLM can be integrated into a closed-loop supervisory controller. The modular architecture supports explainable reasoning, validated execution, and extensibility for future smart building applications.